



Assignment #2

DS-3003

DATA WAREHOUSING & BUSINESS INTELLIGENCE

DEADLINE: Oct 28, 2024, 11pm PST

Total Marks: 100

Weightage: 5%

Important Instructions

- Try to complete the assignment before the deadline.
- Try to attempt all questions! Data warehouse is a highly useful and fast-growing field. This assignment will help you understand the basics well.
- Plagiarism is strictly prohibited! Try to solve the assignment on your own. Marks will be given based on your thought process so make sure you have a solid reason for all your attempts.
- The assignment is of 100 marks in total. Individual marks for each question and part are mentioned below alongside the questions.
- The assignment should be written in a document (MS-word or google-docs) and submitted on Google Classroom (GCR).
- You can use Whimsical, Miro, or Draw.io to create diagrams.

QUESTION: 01 (20 marks)

You are tasked with designing a multidimensional data structure for a data warehousing scenario related to a hotel booking system. The data represents booking information across three dimensions: Room, Time, and Customer. Your goal is to represent this structure in the MDB (Multidimensional Database) format, considering array structures and their linearization.

Array Representation:

- Define the three dimensions (Product, Time, Region) and their respective attributes in a clear and organized manner.
- Represent the array structure in MDB format, indicating the relationships between the dimensions and any measures associated with them.

Linearization:

- **Example Cell:** Choose a specific cell within your multidimensional array, such as the booking count for a specific room type by a particular customer on a certain check-in date.
- **Linearization Process:** Demonstrate the linearization process for this chosen cell by breaking it down step by step using the linearization formula discussed in the material.

Generalization:

- **General Linearization for 'n' Dimensions:** Provide a general formula that represents the linearized index for any cell in the multidimensional structure, applicable for the 'n' dimensions (Room, Time, Customer) in this scenario.

Ensure to provide clear diagrams or illustrations where necessary to enhance your explanation. Additionally, consider how complex queries such as calculating total revenue for all rooms during peak seasons can be managed efficiently through array structures and linearization.

Dataset:

- **Room Dimension:**
RoomID, RoomType, Floor, ViewType
- **Time Dimension:**
Check-in Date, Month, Quarter, Year
- **Customer Dimension:**
CustomerID, CustomerName, MembershipLevel
- **Region Dimension:**
RegionID, RegionName, Country, State, City
- **Measures:**
TotalBookings, Revenue

Example Data:

Room ID	Room Type	Floor	View Type	Check-in Date	Month	Quarter	Year
101	Deluxe	3	Sea view	2024-03-15	March	Q1	2024
102	Suite	5	City view	2024-04-10	April	Q1	2024
103	Standard	2	Park view	2024-05-25	May	Q2	2024
Customer ID	Customer Name	Membership Level	Region ID	Region Name	Country	State	City
01	John Doe	Gold	001	North	USA	NY	NYC
02	Jane Smith	Silver	002	East	UK	ON	Toronto
03	Alex Lee	Bronze	003	West	Canada	CA	LA
Total Bookings	Revenue						
1	300						
2	400						
3	650						

QUESTION: 02 (30 marks)

Suppose that a Data Warehouse for a **University Research Seminar Series** consists of 5 dimensions:

1. **Speakers (Invited to Seminars)**
 - Professors, Industry Experts, PhD Scholars, Entrepreneurs, etc., are types of speakers.

2. **Academic Departments**

- Departments like Computer Science, Data Science, AI, Blockchain, and Research Labs are included.
- Each department has a **Department Head** overseeing activities.

3. **Seminar Details**

- Type of Seminar (Technical Talk, Workshop, Research Discussion).
- Target Audience (Undergraduate, Graduate Students).

4. **Location**

- University Campus locations (Main Campus, City Campus, Research Park).

5. **Time**

- The time dimension can be broken down into **Days, Weeks, Months, Years**.

Measures:

1. Number of Students Attending the Seminar
2. Duration of the Seminar (in hours)

At the Lowest Conceptual Level:

- **Dr. Sarah Ahmed**, (AI Department, Head: Dr. Hashim), presented a **Research Talk** on **March 23, 2023**, at **Main Campus** (Attended by 200 students, lasting **2 hours**).
- **Mr. Ali Khan**, (Blockchain Department, Head: Dr. Khalid), conducted a **Workshop** on **August 19, 2023**, at **City Campus** (Attended by 150 students, lasting **4 hours**).

At higher conceptual levels, the **total number of hours** and **total student attendance** for specific seminars are stored, along with other possible combinations.

Questions:

A) Draw **Snowflake** and **Star Schema** Diagrams for this Data Warehouse scenario. (25 marks)

B) Explain which schema (Snowflake or Star) would be more effective for the **University Research Seminar Series** and justify your reasoning. (05 marks)

QUESTION: 03 (30 marks)

Consider the following scenario for a hotel booking system. The system consists of several hotels, each identified by a code, name, and location. Each hotel offers various room categories, identified by a type-code, description, and price per night. Different types of rooms have a unique room number, floor, bed type, availability status, and view type.

Guests are assigned to rooms and identified by a guest ID, name, contact information, gender, and age. A guest can subscribe to a loyalty program of a specific hotel. The system records details of each guest's room booking, including check-in and check-out dates, room number, and payment information. The system also maintains information about the loyalty points accumulated by each guest for every stay. Loyalty points are recorded as a transaction identified by a transaction number, point amount, and date.

Bookings can be made through travel agencies, each identified by a code, name, address, contact person, and phone number. For every guest's stay, the hotel offers a meal plan. Different meal options are available, identified by a code and description. These meal options include vegetarian, gluten-free, and kosher. For vegetarian and gluten-free meals, the system records the protein level and calorie count, respectively. Kosher meals require approval from the local kosher authority.

1. Draw a Multidimensional ER diagram for the above scenario and identify the fact node and level nodes necessary for analyzing the hotel's operations. (10 marks)
2. Design a Star Schema Diagram for the hotel's Data Warehouse. (10 marks)
3. Design a Snowflake Schema Diagram for the hotel's Data Warehouse. (10 marks)

QUESTION: 04 (20 marks)

Let's consider data evolution of a STELS company that sells digital devices. This company started in Aug 2003 and later in the beginning of 2005, the company decided to build a Data warehouse for better decision-making using company's data. Considering this need, DW for STELS was made on 1st March 2005. STELS initially started production of only mobile phones and launched window 3G smartphones i.e., STELS 65D and STELS 65XD on 1st August 2005. For android users, on 1st February 2006 STELS launched two new phones i.e., STELS GENUS and STELS H11. At the end of 2009, with the increasing demand of STELS products in the market, the product management sector of STELS decided to start the production of digital devices that can be used to browse the internet, watch videos, organize content and much more. Therefore, on 1st October 2010 STELS launched STELS ZOOM Tab. Further 4G started to live up to its hype after 2010. And STELS have made their android phone i.e., STELS GENUS and STELS H11, capable of 4G on 15th January 2012. At the end of 2012, STELS started working on the features of its tablets to grow its business market. So, on 1st August 2012 STELS have introduced two new versions of tablet with additional features i.e., STELS Galaxy Tab and STELS ONE Tab. Later, on 1st March 2013, STELS stopped production of STELS 65D and STELS 65XD and launched metal body smartphones on the same date. The Newly launched mobile phone was STELS U11 and STELS U11+. Consider the above scenario of STELS.

- A. Trace the evolution of data by drawing a Classification Hierarchy Tree. (10 marks)
- B. Construct a Validity Matrix that make sure the correctness of all OLAP queries on STELS Warehouse. (10 marks)

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